

Muhammad Taha Farrukh

+92-333-2758016 | muhammadtahafarrukh@gmail.com | Karachi, Pakistan
github.com/MTahaFarrukh | linkedin.com/in/taha-farrukh

PROFESSIONAL SUMMARY

Computer Science undergraduate (3.97 GPA) specializing in agentic AI and Retrieval-Augmented Generation. Winner, BuildByte 2026 hackathon (IEEE NED UET). Independently built a multi-agent ReAct research pipeline with autonomous web search, scraping, drafting, and self-critique – one of seven solo AI applications shipped in the past year.

EDUCATION

NED University of Engineering and Technology <i>B.S. Computer Science & Information Technology; GPA: 3.97/4.0</i>	Karachi, Pakistan <i>Aug 2024 – May 2028</i>
DHACSS Degree College <i>Intermediate in Computer Science</i>	Karachi, Pakistan <i>Aug 2022 – May 2024</i>

PROJECTS

AI Research Assistant – Multi-Agent Research Pipeline <i>Python, LangChain, Groq, Tavily</i>	GitHub
- Architected a 4-agent ReAct pipeline (Search, Reader, Writer, Critic) using LangChain’s <code>create_react_agent</code> and LCEL, generating full research reports autonomously from a single prompt - Built custom tools for live web search (Tavily API) and content scraping (BeautifulSoup), coordinated via shared-state architecture across agents - Implemented a self-critique loop where a Critic Chain scores and reviews the Writer Chain’s draft, improving output quality without human intervention	
AI Lead Generation Agent – Multi-Agent Lead Generation System <i>Python, Groq, Playwright, FastAPI, React</i>	GitHub
- Architected a 5-agent pipeline (Orchestrator, Prompt, Browser, Email, Export) that converts natural language queries into structured business leads, scraping Google Maps via Playwright and enriching contacts through website crawling - Built Groq-powered intent parsing to translate prompts like “Find coffee shops in Karachi” into structured scraping parameters - Implemented defensive handling for missing phone/email/website data with homepage-first email crawling and contact-page fallback; exposed the pipeline via a FastAPI endpoint with a React dashboard for agent timeline, stats, and sortable results	
CareerGPS AI – Full-Stack AI Career Platform <i>React, FastAPI, Groq, Supabase</i> – 1st Place, BuildByte 2026	GitHub
- Won 1st place among competing teams at BuildByte 2026, a hackathon organized by IEEE NED UET Student Branch - Developed a full-stack AI platform unifying resume analysis, learning roadmaps, job matching, and interview prep into one application - Built a modular FastAPI backend with reusable prompt pipelines across five AI modules; used Supabase for auth/storage, deployed on Vercel	
Virtual Learning System – University Management System <i>C++, Qt, MySQL, OOP</i> – Team Project	
- Built core modules for course creation, student enrollment, and fee management using Qt and C++ with MySQL integration - Designed a relational database and role-based access system (Admin, Teacher, Student) applying OOP principles for scalability	
Additional Projects	
Documind AI – Multimodal RAG system combining Gemini Vision and text embeddings for Q&A over PDF text, images, and diagrams (<i>Python, FastAPI, Gemini Vision</i>) – GitHub City Intelligence AI Agent – HITL, tool-calling agent for weather/news via Tavily & OpenWeather APIs (<i>Python, Streamlit</i>) – GitHub RAG Comparison Platform – Benchmarked 6 chunking/retrieval strategies across independent ChromaDB vector stores (<i>React, FastAPI</i>) – GitHub Nova AI – Personality-driven WhatsApp chatbot with multi-turn memory (<i>Node.js, LangChain, Groq</i>) – GitHub RAG-PDF-Chatbot – Document Q&A app using RAG over user-uploaded PDFs (<i>Streamlit, Groq</i>) – GitHub	

EXTRACURRICULAR, VOLUNTEERING & TEACHING

Volunteer , NED Techfest – Assisted in organizing technical events and supporting event execution (<i>Sept–Oct 2025</i>)
Tutor – Teaching O-Level Computer Science (2210) and Mathematics (4024) to school students
YouTube Educator – Running an educational channel focused on O-Level Mathematics (4024)

HONORS & AWARDS

1st Place, BuildByte 2026 – IEEE NED UET Student Branch Hackathon (Team BuildandDebug – CareerGPS AI)
Merit-Based Academic Scholarship , NED University – Aug 2024 – Present
Merit-Based Laptop Award , Prime Minister of Pakistan
4.0 GPA , 3rd Semester – Dec 2025

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, SQL, HTML, CSS
AI/ML: LangChain, LangGraph, RAG, ReAct Agents, Prompt Engineering, Groq LLMs, Gemini Vision, HuggingFace Embeddings, scikit-learn
Frontend: React, Tailwind CSS, Bootstrap
Backend: FastAPI, Node.js, Express.js
Databases: MySQL, ChromaDB, Supabase, Vector Databases
Tools & Platforms: Git/GitHub, VS Code, Jupyter Notebook, Streamlit, Vercel, Qt, Playwright
Certifications: Python for AI & Data Science (Coursera); Full Stack Development Bootcamp – <i>In Progress</i> (Udemy)
Languages Spoken: English (Fluent), Urdu (Native)